

APPENDIX I CONVERGENCE ANALYSIS OF FCPC

A. Problem Formulation

Consider K clients with dual-skewed non-IID data distributions. The global objective is:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^K p_k F_k(\mathbf{w}) \quad (\text{I.1})$$

where $p_k = N_k / \sum_{i=1}^K N_i$ is the aggregation weight, $F_k(\mathbf{w}) = \mathbb{E}_{(x,y) \sim \mathcal{D}_k} [\ell(\mathbf{w}; x, y)]$ is the local objective, and $\mathcal{D}_k$ represents the dual-skewed non-IID data distribution at client k characterized by both label and quantity skew.

B. Assumptions

We adopt standard assumptions for non-convex federated optimization:

1) **(L-smoothness)** Each local objective F_k is L -smooth:

$$\|\nabla F_k(\mathbf{w}) - \nabla F_k(\mathbf{w}')\| \leq L \|\mathbf{w} - \mathbf{w}'\| \quad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d$$

2) **(Bounded variance)** Stochastic gradient variance at client k :

$$\mathbb{E}_{\xi \sim \mathcal{D}_k} \|\nabla \ell_k(\mathbf{w}; \xi) - \nabla F_k(\mathbf{w})\|^2 \leq \sigma_k^2$$

3) **(Gradient dissimilarity)** The deviation of local gradients is bounded by the heterogeneity metric defined in our framework:

$$\frac{1}{K} \sum_{k=1}^K \|\nabla F_k(\mathbf{w}) - \nabla F(\mathbf{w})\|^2 \leq \zeta^2 + \kappa \cdot \text{JSDN}_{\max}$$

where $\text{JSDN}_{\max} = \max_{i,j} \text{JSDN}_{ij}$ measures the maximum distribution skew in the system.

4) **(Privacy noise bound)** For LDP-perturbed gradients:

$$\mathbb{E} \|\tilde{\nabla} F_k - \nabla F_k\|^2 \leq \sigma_{\text{LDP}}^2 = \frac{dC}{\epsilon^2 N_k} \cdot \Delta f^2$$

with d as the model dimension, C as the class count, and Δf as the sensitivity.

C. Proof of Per-round Contraction

Lemma 1. Under assumptions 1-4, with learning rate $\eta \leq \frac{1}{4L+2\beta}$ and privacy budget $\epsilon > 0$:

$$\begin{aligned} \mathbb{E} \|\mathbf{w}^t - \mathbf{w}^*\|^2 &\leq (1 - \eta \mu_t) \mathbb{E} \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2 + \eta^2 \sum_k p_k (\sigma_k^2 + \sigma_{\text{LDP}}^2) \\ &\quad + 4\eta^2 \beta^2 \mathbb{E} [\Gamma_{\mathcal{P}(t)}] - \eta \beta \sum_{(i,j) \in \mathcal{P}(t)} p_i \mathbb{E} [e^{-\gamma \text{JSDN}_{ij} t}] \end{aligned} \quad (\text{I.2})$$

where $\Gamma_{\mathcal{P}(t)} = \sum_{(i,j) \in \mathcal{P}(t)} p_i \|\nabla F_j(\mathbf{w}^*)\|^2$, $\mu_t = \lambda_{\min}(\nabla^2 F(\mathbf{w}^{t-1}))$, and $\gamma > 0$ is the decay rate.

Proof: **Step 1: Local update decomposition.** For client i paired with j at round t :

$$\mathbf{w}_i^t = \mathbf{w}^{t-1} - \eta \left(\tilde{\nabla} F_i(\mathbf{w}^{t-1}) + 2\beta(\mathbf{w}_i - \mathbf{w}_j^{t-1}) \right) \quad (\text{I.3})$$

$$= \mathbf{w}^{t-1} - \eta \mathbf{g}_i^t \quad (\text{I.4})$$

where $\mathbb{E} \tilde{\nabla} F_i = \nabla F_i$ and $\mathbb{E} \|\tilde{\nabla} F_i - \nabla F_i\|^2 \leq \sigma_{\text{LDP}}^2$.

Step 2: Global update aggregation.

$$\mathbf{w}^t = \sum_{i=1}^K p_i \mathbf{w}_i^t = \mathbf{w}^{t-1} - \eta \mathbf{g}_t \quad (\text{I.5})$$

$$\mathbf{g}_t = \sum_{i=1}^K p_i \mathbf{g}_i^t = \sum_{i=1}^K p_i \tilde{\nabla} F_i + 2\beta \sum_{(i,j) \in \mathcal{P}(t)} p_i (\mathbf{w}_i - \mathbf{w}_j^{t-1}) \quad (\text{I.6})$$

Step 3: Deviation from optimum.

$$\begin{aligned} \|\mathbf{w}^t - \mathbf{w}^*\|^2 &= \|\mathbf{w}^{t-1} - \eta \mathbf{g}_t - \mathbf{w}^*\|^2 \\ &= \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2 - 2\eta \langle \mathbf{g}_t, \mathbf{w}^{t-1} - \mathbf{w}^* \rangle + \eta^2 \|\mathbf{g}_t\|^2 \end{aligned} \quad (\text{I.7})$$

Step 4: Expectation of inner product. Using the strong convexity property near the optimum:

$$\begin{aligned}\mathbb{E}\langle \mathbf{g}_t, \mathbf{w}^{t-1} - \mathbf{w}^* \rangle &= \mathbb{E}\langle \nabla F(\mathbf{w}^{t-1}), \mathbf{w}^{t-1} - \mathbf{w}^* \rangle \\ &\quad + 2\beta \sum_{(i,j) \in \mathcal{P}(t)} p_i \mathbb{E}\langle \mathbf{w}_i - \mathbf{w}_j^{t-1}, \mathbf{w}^{t-1} - \mathbf{w}^* \rangle\end{aligned}\quad (\text{I.8})$$

$$\geq \mu_t \|\mathbf{w}^{t-1} - \mathbf{w}^*\|^2 + \frac{\beta}{2} \sum_{(i,j) \in \mathcal{P}(t)} p_i e^{-\gamma \text{JSDN}_{ij} t} \quad (\text{I.9})$$

The exponential term arises from the mutual constraint mitigating dimensional collapse, where the rate depends on the dissimilarity JSDN_{ij} .

Step 5: Expectation of gradient norm.

$$\mathbb{E}\|\mathbf{g}_t\|^2 \leq 2\mathbb{E}\left\|\sum_i p_i \tilde{\nabla} F_i\right\|^2 + 8\beta^2 \mathbb{E}\left\|\sum_{(i,j) \in \mathcal{P}(t)} p_i (\mathbf{w}_i - \mathbf{w}_j^{t-1})\right\|^2 \quad (\text{I.10})$$

$$\leq 2 \sum_i p_i^2 (\sigma_i^2 + \sigma_{\text{LDP}}^2) + 2\zeta^2 + 8\beta^2 \Gamma_{\mathcal{P}(t)} \quad (\text{I.11})$$

Step 6: Final assembly. Combining the bounds and assuming $\eta \leq 1/(4L + 2\beta)$, we obtain the final contraction result in Eq. I.2.

D. Full Proof of Main Convergence Theorem

Theorem 1. After T communication rounds with $\eta = \frac{\mu_0}{\sqrt{T}(\Sigma + \beta^2 \Gamma_{\max})}$ and regularization $\beta = O(1/T^{1/4})$:

$$\begin{aligned}\min_{0 \leq t \leq T} \mathbb{E}\|\nabla F(\mathbf{w}^t)\|^2 &\leq \frac{2\Delta_F}{\eta T} + \frac{\eta L}{K} (\sigma^2 + \sigma_{\text{LDP}}^2) + 4\eta^2 L^2 \zeta^2 \\ &\quad + 4\eta^2 L^2 \kappa \beta^2 \overline{\text{JSDN}} - C\eta^2 L^2 \sum_{t=1}^T e^{-\gamma \text{JSDN}_{\mathcal{P}(t)} t}\end{aligned}\quad (\text{I.12})$$

where $\Delta_F = F(\mathbf{w}^0) - F^*$ is initial error, σ^2 is the aggregated variance, and $\overline{\text{JSDN}} = \frac{1}{T} \sum_{t=1}^T \text{JSDN}_{\mathcal{P}(t)}$.

Proof: Summing Lemma 1 over $t = 0$ to $T - 1$ and rearranging terms:

$$\sum_{t=0}^{T-1} \mathbb{E}\|\mathbf{w}^t - \mathbf{w}^*\|^2 \leq \frac{\mathbb{E}\|\mathbf{w}^0 - \mathbf{w}^*\|^2}{\eta \mu_0} + \frac{\eta T \Sigma}{\mu_0} - \frac{\beta C_0}{\mu_0} \sum_{t=1}^T e^{-\gamma \overline{\text{JSDN}} t} \quad (\text{I.13})$$

Using the inequality $\mathbb{E}\|\nabla F(\mathbf{w}^t)\|^2 \leq L\mathbb{E}\|\mathbf{w}^t - \mathbf{w}^*\|^2$ and taking the minimum over t :

$$\min_{0 \leq t \leq T} \mathbb{E}\|\nabla F(\mathbf{w}^t)\|^2 \leq \frac{L}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\mathbf{w}^t - \mathbf{w}^*\|^2 \quad (\text{I.14})$$

Substituting the appropriate learning rate η^* and β^* yields the final bound, demonstrating that the convergence rate is modulated by the JSDN pairing efficiency.

APPENDIX II

THEORETICAL JUSTIFICATION FOR JSDN AND PARAMETER ROBUSTNESS

In this section, we provide the theoretical basis for the proposed Jensen-Shannon Divergence and Number (JSDN) index and analyze the rationality of the hyperparameter λ (set to 0.3 in our experiments). We demonstrate that JSDN serves as a quantifiable proxy for the upper bound of weight divergence caused by data heterogeneity.

A. Gradient Divergence Upper Bound

Let $F_i(\mathbf{w})$ be the local loss function for client i . In Federated Learning, the heterogeneity between clients i and j leads to "Client Drift," quantified by the difference in their expected gradients.

Following standard assumptions in non-IID analysis (e.g., Li et al., ICLR 2020), the gradient divergence is bounded by two distinct sources: distribution discrepancy and sampling variance. We formally model the upper bound of the gradient difference between client i and client j as:

$$\|\nabla F_i(\mathbf{w}) - \nabla F_j(\mathbf{w})\| \leq \underbrace{A \cdot \mathcal{D}(P_i(y), P_j(y))}_{\text{Distribution Shift}} + B \cdot \underbrace{\left| \frac{1}{\sqrt{N_i}} - \frac{1}{\sqrt{N_j}} \right|}_{\text{Sampling Variance}} \quad (\text{II.1})$$

where:

- $\mathcal{D}(P_i, P_j)$ measures the distance between label distributions (Label Skew).
- N_i, N_j represents the sample sizes (Quantity Skew), which affect the variance of the stochastic gradient estimate.
- A and B are constants related to the model's Lipschitz smoothness and data complexity.

B. Mapping JSDN to Gradient Divergence

The proposed JSDN metric is designed to capture these two heterogeneity sources.

- We utilize Jensen-Shannon Divergence ($\mathcal{JSD}$) to quantify $\mathcal{D}(P_i, P_j)$.
- We utilize the normalized quantity difference $N_{diff} = \frac{|N_i - N_j|}{N_i + N_j}$ to proxy the sampling variance term.

Substituting these into Eq. (II.1), the heterogeneity metric H_{ij} can be approximated as a linear combination:

$$H_{ij} \approx c_1 \cdot \mathcal{JSD}_{ij} + c_2 \cdot N_{diff} \quad (\text{II.2})$$

Comparing this to our JSDN formulation:

$$\text{JSDN}_{ij} = (1 - \lambda) \cdot \mathcal{JSD}_{ij} + \lambda \cdot N_{diff} \quad (\text{II.3})$$

It becomes evident that λ represents the relative contribution of quantity skew to the total gradient divergence compared to label skew.

C. Rationality of $\lambda = 0.3$

The choice of $\lambda = 0.3$ (implying $1 - \lambda = 0.7$) is theoretically grounded in the nature of neural network optimization:

- 1) **(Sensitivity Analysis)** Label distribution skew changes the geometry of the loss landscape (altering the optimal global solution $\mathbf{w}^*$), creating a *directional bias* in updates. Quantity skew, conversely, primarily affects the *magnitude stability* and convergence speed but shares the same optimization objective.
- 2) **(Dominance Principle)** Since directional correctness is a prerequisite for convergence, the Distribution Shift term (A) typically dominates the Sampling Variance term (B) in non-IID settings. Therefore, theoretically, we must have $(1 - \lambda) > \lambda$, i.e., $\lambda < 0.5$.
- 3) **(Conclusion)** $\lambda = 0.3$ is a robust empirical approximation of the ratio $\frac{B}{A+B}$, prioritizing the correction of semantic divergence (label skew) while maintaining regularization for quantity imbalance.

D. Adaptive Capability via Limit Analysis

A key concern is whether a fixed λ applies to extreme scenarios. We define the pairing strategy as finding the pair (i, j) that maximizes JSDN: $\arg \max_{(i,j)} \text{JSDN}_{ij}$. We analyze the limits:

Case 1: Extreme Label Skew (No Quantity Skew). Let $N_i \approx N_j$, then $N_{diff} \rightarrow 0$.

$$\lim_{N_{diff} \rightarrow 0} \text{JSDN}_{ij} = (1 - \lambda) \mathcal{JSD}_{ij} \quad (\text{II.4})$$

Since $(1 - \lambda)$ is a positive constant, maximizing JSDN is mathematically equivalent to maximizing $\mathcal{JSD}$. The pairing logic automatically adapts to focus purely on label heterogeneity.

Case 2: Extreme Quantity Skew (No Label Skew). Let $P_i \approx P_j$, then $\mathcal{JSD} \rightarrow 0$.

$$\lim_{\mathcal{JSD} \rightarrow 0} \text{JSDN}_{ij} = \lambda N_{diff} \quad (\text{II.5})$$

Similarly, maximizing JSDN becomes equivalent to maximizing quantity difference. The algorithm adapts to pair small-data clients with large-data clients to leverage stability transfer.

System Evaluation: This limit analysis proves that FCPC does not require dynamic tuning of λ to handle single-skew extremes. The linear formulation acts as a *soft gating mechanism*, where the dominant heterogeneity source naturally dictates the pairing logic. The value $\lambda = 0.3$ only governs the trade-off in the specific "Dual Skew" zone, ensuring the framework remains robust across the entire spectrum of non-IID scenarios.